

Project Report: E-Commerce Sales Analysis



Executive Summary

This report presents an analysis of ecommerce sales data using Power BI. It highlights regional performance, category trends, payment preferences, and profitability challenges.

The insights and recommendations aim to help management optimize inventory, pricing, and customer engagement strategies.

Objective

To analyze sales performance across regions, categories, and customer segments for an e-commerce business using Power BI.

Tools & Methodology

- Power BI for dashboard creation
- CSV datasets (Details.csv, Orders.csv)
- Data cleaning and modeling
- DAX measures for KPIs (Amount, Profit, Quantity, Average Order Value)



Key Insights

- **Revenue Leader:** Maharashtra contributed the highest sales amount.
- **Top Category:** Clothing accounted for 63% of total quantity sold.
- **Payment Preference:** COD was the most used payment mode (44%), followed by UPI and Credit Card.
- **Profit Volatility:** Negative profits observed in April, May, June, and September—indicating seasonal challenges or discount periods.
- **Customer Value:** Harivansh and Madhav were high-value customers.
- **Sub-Category Performance:**
 - High Profit: Printers, Accessories
 - Loss-Making: Sarees and Bookcases showed frequent negative profits



Recommendations

- **Optimize Inventory:** Focus on high-performing sub-categories like Printers and Accessories.
- **Revisit Pricing Strategy:** Investigate losses in Saree and Bookcase segments.
- **Promote Digital Payments:** Encourage UPI and Credit Card usage to reduce COD dependency.
- **Seasonal Planning:** Prepare for low-profit months with targeted campaigns or cost control.



Business Impact

- Provides clarity on regional sales gaps and customer value.

- Helps management identify profitable categories and reduce losses.
- Supports digital payment adoption, reducing COD risks.
- Enables proactive planning for seasonal profit dips.

Conclusion

The ecommerce dashboard provides a clear view of sales and profitability drivers.

By focusing on profitable sub-categories, encouraging digital payments, and planning for seasonal dips, the business can achieve sustainable growth.